Final Review: Practice Problems

Number Theory

Problem 1

In each of the following, find a value of x that satisfies the given congruence, or argue that no such x exists. Show how you found x.

a. $4(x-3) \equiv 8x-3 \pmod{41}$

b. $7(x+5) \equiv 2x \pmod{13}$

Problem 2

The value of the *Euler ϕ function* at the positive integer n is defined to be the number of positive integers less than or equal to n that are relatively prime to n.

a. Find the following values:

(i) $\phi(8)$

(ii) $\phi(7)$

(iii) $\phi(15)$

b. Prove that n is composite iff $\phi(n) < n-1$.

Problem 3

Suppose $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$. Let $n \in \mathbb{Z}^+$. Prove the following statements:

a. $ac \equiv bd \pmod{m}$.

b. $a^n \equiv b^n \pmod{m}$.

Hint: Try induction on n.

Counting

Problem 1

Consider a finite set S. Count the number of pairs of non-overlapping subsets, justifying your answer.

Note: The order in which you pick the pair does not matter.

Problem 2

Consider n distinct objects. How many distinct sequences are there of these n non-repeating items (any length)?

Problem 3

How many different ways are there to list the numbers $1, 2, \ldots, 2n$ such that the even numbers appear in increasing order and the odd numbers appear in decreasing order?

Probability

Problem 1

Consider a line of n aliens at a spaceport looking to board a rocket containing n free seats. Each of them has a ticket, and the i^{th} alien is assigned to the i^{th} seat.

The first alien in line enters the rocket, and instead of looking at its ticket for its assigned seat, sits uniformly at random. Every following alien that enters chooses its assigned seat if it is available, otherwise chooses one of the remaining seats uniformly at random. What is the probability that the n^{th} alien sits down in its assigned seat?

Problem 2

You have 100 coins on a table. 99 of them are fair and 1 of them has H on both sides.

a. You choose a coin uniformly at random, and flip it. Let X be a random variable where $X = 1$ if you get heads, and $X = 0$ if you get tails. What is $E[X]$?

b. You choose a coin again uniformly at random, flip it, record the result, and put the coin back. You repeat this experiment 9 more times (for a total of 10). Let X be a random variable denoting the total number of heads you get. What is $E[X]$?

c. You choose a coin again uniformly at random, but flip this *same coin* 10 times. If you get heads every time, what is the probability you will also get heads the 11th time?

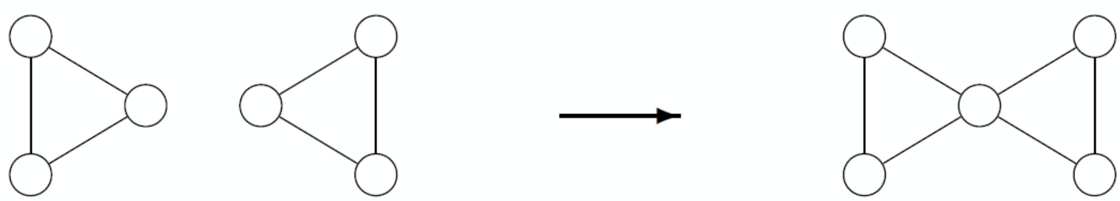
Graph Theory

Problem 1

Show that, in any graph G, if there is a vertex v of odd degree, then there is a path from v to some other vertex of odd degree.

Problem 2

Prove that by joining together any $n \geq 2$ connected graphs that have connected Eulerian tours the resulting graph will also have a Eulerian tour. In this problem, we define joining as merging any two vertices together into a single new vertex, as depicted in the image below:



Problem 3

Pluto spotted this proof of the 'Graph Pigeonhole Principle.'

Claim: For $n \geq 4$, if an n vertex graph has at least $n + 1$ edges, it must contain a subgraph that is a triangle (a simple cycle of length 3).

Proof:

Define $P(n)$: For every graph on n vertices, if the graph has at least $n+1$ edges, it contains a subgraph that is a triangle.

Consider the base case $P(4)$. The complete graph K_4 has 6 edges and vertices $\{v_1, v_2, v_3, v_4\}$. We're interested in a graph on these vertices with at least $n+1=5$ edges. If the graph has 6 edges, it is a complete graph and any 3 of its vertices forms a triangle. If it has 5 edges, we can assume without loss of generality that the missing edge is (v_1, v_2). The edges $\{v_1, v_3\}$, $\{v_3, v_4\}$, and $\{v_1, v_4\}$ are in the graph and form a triangle. Hence, $P(4)$ is true, which proves our base case.

Assume that $P(k)$ is true for $n = k$. For the inductive step, consider a graph G with k vertices and at least $k+1$ edges. By our inductive hypothesis, G contains a triangle. Construct an $k+1$ vertex graph G' by adding one vertex to G and at least one edge (so the number of edges is at least $k+2$). By the inductive hypothesis, G contains a triangle. Since G is a subgraph of G', G' also contains a triangle. Hence, if $P(k)$ is true, so is $P(k+1)$.

We have shown that $P(4)$ holds and that if $P(k)$ holds so does $P(k+1)$. Thus, we have shown that $P(n)$ holds for all $n \geq 4$ by the principle of induction. QED.

a. Prove that Pluto's Graph Pigeonhole Principle is false.

b. What's wrong with Pluto's proof?

c. Repair the proof by showing that a simple graph with n vertices where the minimum degree is more than $n/2$ must contain a triangle.

   Hint: Pigeonhole Principle!